Behind the Curtain: GDPR Literacy and the Privacy Paradox

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Abstract

This paper tests whether individual-level awareness of the General Data Protection Regulation (GDPR) translates into privacy-protective behavior among European citizens, and whether this relationship is mediated by perceived control over personal data. Using Special Eurobarometer 487a (EB 91.2, 2019, N = 10,263) and weighted binary logistic regression estimated via maximum likelihood, we find strong support for a positive effect: hearing of GDPR increases the probability of changing privacy settings by 14.0 percentage points, and each additional data right known adds a further 6.0 percentage points, while stated concern about data use is negatively associated with action, replicating the privacy paradox. Formal mediation analysis (5,000 bootstrap simulations) shows that perceived control accounts for only 3.0% of the awareness–behavior relationship, suggesting that regulatory literacy motivates protective action through direct pathways rather than enhanced felt control. The findings suggest that part of the governance gap may reflect limits in regulatory communication.

Keywords: GDPR; privacy paradox; privacy-protective behavior; regulatory awareness; data rights; EU data protection

I. Introduction

"Pay no attention to that man behind the curtain!"

— THE WIZARD OF OZ (1939)

Much like the Wizard's command to ignore the machinery of Oz, the surveillance economy thrives on consumer opacity. The General Data Protection Regulation (GDPR), adopted by the European Union in 2016, was designed to pull back this curtain. Replacing the 1995 Data Protection Directive, the GDPR's primary objectives are to protect the personal data of individuals in the European Economic Area (EEA), harmonize laws, and strengthen individual privacy rights. Article 4 of the GDPR defines personal data as any "information relating to an identified or identifiable natural person." As of 2025, with over six billion people online generating upwards of 400 million terabytes of data daily, the protection of this data has transitioned from a technical niche to a fundamental pillar of the Charter of Fundamental Rights of the EU (International Telecommunications Union, 2025 ; Statista, 2024).

However, the formal strength of these rights is often dwarfed by the practical reality of data exploitation. While the 2018 Cambridge Analytica scandal, which saw 87 million Facebook users' data leveraged to influence democratic elections, exposed a massive power imbalance between corporations and citizens, the regulatory response remains fundamentally passive (Cadwalladr & Graham-Harrison, 2018). Privacy, in this context, is a struggle for balance. As the data is the oil of this digital economy, companies have a structural incentive to leverage it; users, meanwhile, are granted legal rights that they must proactively exercise to protect their privacy.

This raises a vital concern regarding the "execution" of these rights. If users do not, or do not know how, execute their rights, does the protection exist in anything but name? Seven years into the enforcement of the GDPR and the subsequent AI Act, the discrepancy between legal protection and actual privacy continues to intensify alongside the "capitalization of life without limit" (Couldry & Mejias, 2019).

This paper investigates whether the implementation of the GDPR and the specific level of awareness among citizens at a country level actually translates into privacy-protective behaviors. If awareness does not lead to action, it suggests a systemic failure: perhaps we have established the legal rights to privacy, but we lack the practical tools required to execute them.

II. Theory

The theoretical backdrop for understanding consumer data decisions is the "privacy paradox", defined as the persistent inconsistency between individuals' high stated privacy concerns and their actual data-disclosing behavior (Brown, 2001; Norberg et al., 2007). Behavioral economics and psychology literature suggest this attitude-behavior gap stems from bounded rationality, incomplete information, and cognitive biases (Acquisti (2004)). Because of psychological deviations like "hyperbolic discounting", individuals systematically trade their long-term privacy for immediate, short-term gratification (Acquisti, 2004).

Empirically, while up to 89.2% of users report high privacy concerns, they frequently release sensitive personal data, such as social security numbers, for small financial rewards or convenience (Acquisti, 2004). In social contexts, the gap is exacerbated as users prioritize the accumulation of "social capital" and emotional rewards over machine-readable data protection (Engels, 2019). Notably, granting users a greater ability to manage their data can sometimes trigger the “paradox of control”, where the illusion of empowerment increases self-efficacy but paradoxically decreases attention to privacy, leading to greater data disclosure (Brandimarte, Acquisti and Loewenstein, 2013).

Recent empirical studies examining the GDPR and analogous state-level policies reveal complex behavioral responses. On one hand, clear regulatory mechanisms can prompt protective action: following GDPR's implementation, mandated explicit opt-out mechanisms resulted in a 12.5% drop in observed consumer cookies (Aridor et al., 2024). This suggests that regulatory interventions can successfully motivate privacy-conscious users to utilize explicit legal opt-outs (Aridor et al., 2024). On the other hand, robust privacy policies can also inadvertently reduce protective actions. Recent evidence shows that strong government regulations and corporate privacy policies significantly increase consumers' digital trust and reduce expressed privacy concerns (Demirici et al., 2025). Because consumers feel safely protected by the regulatory environment, they actually engage in more data sharing and reduce defensive privacy behaviors (Demirici et al., 2025).

Despite this growing body of empirical work, a critical gap remains regarding the specific psychological mechanisms through which regulatory awareness translates into proactive privacy management. Prior literature has predominantly modeled GDPR awareness as a dependent variable (an outcome of digital experience) (Rughinis et al., 2021), or focused heavily on “digital trust” and “reduced concern” as the primary mediators between regulation and consumer behavior (Ajzen, l. 2002 ; Mubarak & Petraite, 2020; Tunkevichus & Rebiazina, 2021). Consequently, it remains untested whether individual-level GDPR awareness acts as an independent catalyst driving protective behavior.

This paper fills this gap by testing a forward causal chain: whether GDPR awareness instills a sense of perceived control, which in turn drives privacy-protective behavior. Based on the conflicting evidence in prior work, this literature generates two competing directional hypotheses:

- **Hypothesis 1** (Positive Effect): GDPR awareness has a positive effect on privacy-protective behavior, driven by an increased sense of perceived control that empowers users to exercise their newly salient rights.
- **Hypothesis 2** (Null/Negative Effect): GDPR awareness has a null or negative effect on privacy-protective behavior, because awareness of strong regulations generates institutional trust and reduces privacy concerns, thereby diminishing the user's perceived need to engage in defensive actions.

III. Methodology

A. Data

This paper utilizes two waves of the Special Eurobarometer survey series administered by the European Commission. The primary dataset is Special Eurobarometer 487a (EB 91.2), conducted in March 2019 across all 28 EU member states (N = 27,524). The latter is the first dedicated EU-wide data protection survey conducted after GDPR implementation in May 2018, making it suited to examine post-regulatory awareness and behavior. Interviews were conducted face-to-face in respondents' native languages using a multi-stage random probability sampling design. Post-stratification weights (*w1*) are applied throughout all analyses to account for sampling design and ensure national representability.

The secondary dataset is Special Eurobarometer 431 (EB 83.1), conducted in 2015 prior to GDPR implementation ($N \approx 27{,}000$). This wave is used not as a comparative cross-section but to construct a country-level contextual predictor, the mean pre-GDPR felt control over online data disclosure (question B4), which is aggregated at the country level and merged into the 2019 individual-level dataset. This allows to account for the nested structure of the data, in which individuals are embedded within national privacy cultures that preceded the regulation and responds to the concern that observed individual-level associations may reflect pre-existing country-level variation rather than regulatory effects.

The final dataset from EB 91.2 after deletion of missing values on all model variables is N = 10,263, this reduction is due primarily to several non-response on the behavioral dependent variable. This reduction is noted as a limitation; however, the retained sample remains large and geographically representative across all 28 member states (See Table A1 in the Appendix).

B. Variables

The **dependent variable** is privacy-protective behavior (question B11). The primary outcome variable is a binary indicator derived from QB11: "Have you ever tried to change the privacy settings of your personal profile on a social network or online platform?" Respondents answering "yes" are coded 1 and those answering "no" are coded 0. This variable accounts for the behavioral dimension of the privacy paradox: the gap between stated concern and actual protective action. Changing privacy settings is chosen as the dependent variable because it is a direct, low-friction behavioral act that GDPR explicitly empowers citizens to perform, and because it represents the type of proactive rights-exercise the regulation was designed to encourage.

The **mediator** is the perceived control (question B9). The mediating variable is derived from question B9: "How much control do you feel you have over the information you provide online?" The survey data scale ranges from 1 (a lot of control) to 4 (no control at all) and is reversed prior to analysis so that higher values indicate greater felt control (1 = no control, 4 = a lot of control).

Following a Brant test that rejected the proportional odds assumption for an ordered logit specification (Omnibus $\chi^2 = 46.49$, df = 4, $p < 0.001$; see Table A2 in the Appendix), driven by a violation for the concern-about-data-use predictor, question B9 is transformed into a binary indicator: respondents answering “some” or “a lot” of control are coded 1 (71.5% of the reviewed sample used); those answering "not much" or “no” control are coded 0 (28.5%).

The **independent variable** is the GDPR awareness (question B17, question B18). This variable encompasses two dimensions of GDPR awareness, following the theoretical distinction between regulatory exposure and regulatory literacy. Question B17 (“Have you heard of the GDPR?”) is coded as an indicator (1 = yes, 0 = no), capturing passive exposure to the regulation. Questions B18_1 through B18_6 ask respondents whether they have heard of each of six specific rights granted under GDPR: the right to access, correct, delete, and port personal data, the right to object to direct marketing, and the right to not be subject to automated decisions. Each item (data right) is coded 1 if the respondent has heard of that right and 0 otherwise, these are later summed into a knowledge score ranging from 0 to 6. The latter illustrates regulatory literacy, meaning knowing what the regulation actually grants.

The two awareness measures are treated as distinct predictors; their low inter-correlation ($r = 0.20$) confirms that regulatory exposure and knowledge of data rights capture empirically separable dimensions of GDPR awareness (see Table A3 in the Appendix).

The **control variables**. Two individual-level controls are included: question B10 (“To what extent do you feel that the use of your personal data by companies is a concern for you?”) captures concern about commercial data use and question B7 (“To what extent are you concerned about online tracking?”) captures concern about behavioral surveillance. Both are retained in their original ordinal form. These controls are motivated by the privacy paradox literature: if concern alone drives behavior independently of GDPR awareness, failing to include it would bias the awareness coefficients.

The **country-level contextual predictor.** The country-level mean of question B4 from EB 83.1 (2015) - measuring pre-GDPR felt control over online data disclosure, is included as a contextual predictor. This variable accounts for heterogeneity in national privacy cultures that preceded the regulation and provides analytical use of the 2015 wave beyond descriptive comparison.

C. Descriptive Statistics

Table 1 presents descriptive statistics for all key variables in the analytic sample.

Table 1. Descriptive Statistics, EB 91.2 (2019)

Variable	Mean	SD	Min	Max	*N*
Changed settings (QB11)	0.61	0.49	0.00	1.00	13,048
Perceived control (QB9, binary)	0.71	0.45	0.00	1.00	13,048
GDPR heard (QB17)	0.48	0.50	0.00	1.00	13,048
Knowledge score (QB18, 0–6)	1.32	1.78	0.00	6.00	13,048
Concern: data use (QB10)	2.33	0.79	1.00	4.00	10,690
Concern: tracking (QB7)	4.38	1.16	1.00	6.00	13,034
Country baseline control (2015)	2.16	0.14	1.97	2.64	12,552

Source: Special Eurobarometer 487a (EB 91.2, 2019). Post-stratification weights applied. Sample: N = 10,263 (after missing values deletion).

Notable descriptive patterns include the right-skewed distribution of the knowledge score: 50.7% of respondents know zero GDPR rights, while 5.5% know all six. Among those who have heard of GDPR (QB17 = 1), 40.7% are unable to name a single right guaranteed by the regulation.

D. Model Specification

The paper employs binary logistic regression estimated via maximum likelihood, which is appropriate given the binary nature of both the primary dependent variable (question B11) and the mediating variable (question B9). Binary logit models the log-odds of the outcome as a linear function of the predictors, avoiding the heteroskedasticity and out-of-bounds prediction problems of the linear probability model while producing interpretable coefficients via transformation to odds ratios or average marginal effects (Long, 1997).

Three models are estimated to decompose the total effect of GDPR awareness into direct and mediated components:

Model 1 (M1, Eq. 1) estimates the total effect of GDPR awareness on privacy-protective behavior:

$$log[P(Y_i=1) / (1-P(Y_i=1))] = \beta_0 + \beta_1 Heard_i + \beta_2 Score_i + \beta_3 ConcUse_i + \beta_4 ConcTrack_i + \beta_5 BaseCtrl_c + \varepsilon_i \quad (1)$$

Model 2 (M2, Eq. 2) estimates the effect on the mediator:

$$log[P(M_i=1) / (1-P(M_i=1))] = \alpha_0 + \alpha_1 Heard_i + \alpha_2 Score_i + \alpha_3 ConcUse_i + \alpha_4 ConcTrack_i + \alpha_5 BaseCtrl_c + \varepsilon_i \quad (2)$$

Model 3 (M3, Eq. 3) adds perceived control to isolate the direct effect:

$$log[P(Y_i=1) / (1-P(Y_i=1))] = \gamma_0 + \gamma_1 Heard_i + \gamma_2 Score_i + \gamma_3 Ctrl_i + \gamma_4 ConcUse_i + \gamma_5 ConcTrack_i + \gamma_6 BaseCtrl_c + \varepsilon_i \quad (3)$$

where i indexes individuals and c indexes countries. All models are estimated with Eurobarometer post-stratification weights. Results are reported as log-odds coefficients and average marginal effects (AMEs), which express the change in the predicted probability of the outcome associated with a unit change in each predictor, averaged across all observations.

A formal mediation test is conducted using the mediation package in R, employing nonparametric bootstrapping with 5,000 simulations to decompose the total effect into the Average Causal Mediation Effect (ACME, the indirect path through perceived control) and the Average Direct Effect (ADE). The proportion of the total effect mediated by question B9 is reported for each awareness dimension. Given the fact that in a cross-sectional dataset we cannot check the key assumption needed to prove mediation is truly causal; mediation results are interpreted as descriptive decompositions of the total association rather than causally identified indirect effects.

E. Model Validity and Specification

Functional form. The binary logit is the appropriate functional form for a binary outcome, satisfying the bounded probability constraint that the linear probability model violates. The proportional odds assumption was tested and rejected for the ordered specification of question B9, justifying the binary collapse of the mediating variable.

Omitted variable bias. The primary threat to validity is unobserved individual-level confounding: respondents who are both GDPR-aware and likely to change settings may share underlying characteristics – such as general digital literacy, institutional trust, or pre-existing privacy disposition, are not fully captured by the included controls. The country-level baseline from EB 83.1 partially addresses national-level confounding, but individual-level omitted variables cannot be ruled out in a cross-sectional design. Relatedly, reverse causality cannot be excluded: individuals who changed settings may have subsequently sought out information about GDPR, inverting the assumed causal direction. These limitations are acknowledged; the observed associations are interpreted as conditional correlations rather than causal effects.

Multicollinearity. The two GDPR awareness measures (questions B17 and B18) are weakly correlated ($r = 0.20$) and are included simultaneously without risk of collinearity inflation. Variance inflation factors are below 2.0 for all predictors.

Robustness. Three robustness checks are reported: replacing the knowledge score with an indicator (knows at least one GDPR right), substituting the country-level baseline from pre-GDPR perceived control (question B4) to pre-GDPR privacy concern (QB5), and excluding respondents with a perfect knowledge score of 6 to test whether results are driven by a small expert cluster.

IV. Analyses and Results

A. Total effects of GDPR awareness (M1)

Table 2 reports log-odds coefficients for all three models. Figure 1 visualises the full coefficient estimates with 95% confidence intervals.

Table 2. Logistic regression models: GDPR awareness and privacy-protective behavior

	M1: Total Effect **(DV: Changed Settings)**	**M2: Mediator** **(DV: Control)**	**M3: Full Model** **(DV: Changed Settings)**
(Intercept)	2.807***	4.614***	2.001***
	(0.334)	(0.327)	(0.342)
GDPR heard (QB17)	0.637***	0.095*	0.633***
	(0.042)	(0.042)	(0.042)
Knowledge score (QB18)	0.277***	0.074***	0.270***
	(0.015)	(0.013)	(0.015)
Concern: data use (QB10)	−0.169***	0.076**	−0.180***

	(0.026)	(0.026)	(0.026)
Concern: tracking (QB7)	−0.266***	−0.155***	−0.251***
	(0.020)	(0.020)	(0.020)
Country baseline control (2015)	−0.629***	−1.641***	−0.433**
	(0.140)	(0.137)	(0.142)
Perceived control (QB9)			0.538***
			(0.043)
N	10,263	10,263	10,263
AIC	14,318.3	14,511.1	14,158.2
BIC	14,361.7	14,554.5	14,208.9
Log-likelihood	−7153.1	−7249.6	−7072.1
RMSE	0.47	0.47	0.47

Note: Weighted binary logistic regression. Coefficients are log-odds. Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

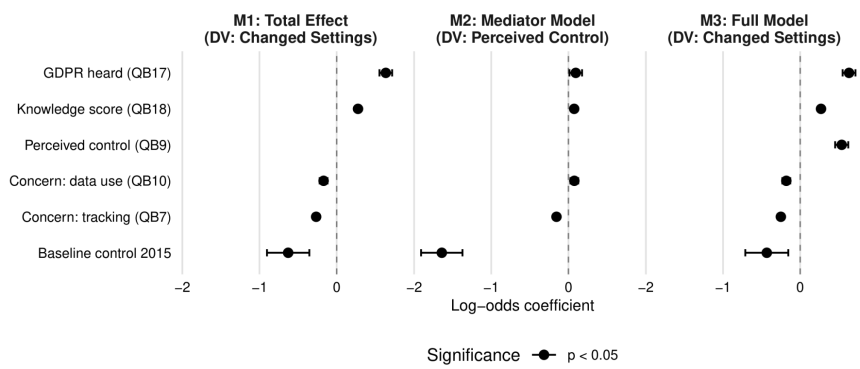

Figure 1. Logistic regression coefficients (log-odds) with 95% CIs across M1, M2, and M3. Filled circles indicate $p < 0.05$; open circles $p \geq 0.05$.

Model 1 provides evidence consistent with Hypothesis 1. Both dimensions of GDPR awareness are positive, large, and highly significant predictors of privacy-protective behavior. Individuals who have heard of GDPR are substantially more likely to have changed their privacy settings ($\widehat{\beta} = 0.637$, $p < 0.001$), corresponding to an AME of 14.0 percentage points (95% CI [12.2, 15.8]). Each additional GDPR right known is associated with a 6.0 percentage point increase in this probability (AME = 0.060, 95% CI [0.054, 0.066], $p < 0.001$), such that full rights knowledge (score = 6) relative to no knowledge predicts a 36 percentage point difference in behavioral engagement.

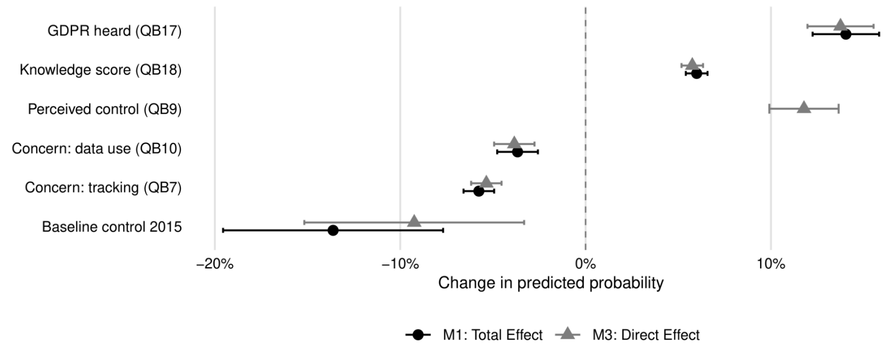

Figure 2. Predicted probability of having changed privacy settings by rights knowledge score and GDPR exposure. All other predictors held at sample means. Shaded bands = 95% CIs.

Notably, the two concern variables are both negative and statistically significant. Concern about commercial data use (AME = - 0.037, $p < 0.001$) and concern about online tracking (AME = - 0.058, $p<0.001$) are associated with lower probabilities of settings-changing behavior – a direct empirical manifestation of the privacy paradox.

The country-level baseline predictor is negative and significant ($\widehat{\beta}$ = - 0.629, $p<0.001$; AME = - 0.136). Countries where citizens already felt more in control in 2015 show lower rates of settings-adjustment in 2019, consistent with a ceiling effect.

B. Effect of GDPR awareness on perceived control (M2)

Model 2 tests whether GDPR awareness shifts citizens' sense of felt control. Both awareness measures reach statistical significance: hearing of GDPR raises the probability of reporting felt control by 2.1 percentage points (AME = 0.021, 95% CI [0.003, 0.039], $p = 0.023$), and each additional right known adds 1.6 percentage points (AME = 0.016, $p < 0.001$). However, these effects are approximately seven times smaller than the corresponding effects on behavior in M1. The country-level baseline exerts the strongest effect in this model (AME = - 0.360, $p < 0.001$).

C. Direct effects and mediation (M3)

Model 3 adds perceived control (QB9) alongside GDPR awareness. Perceived control is independently associated with an 11.8 percentage point increase in the probability of having changed settings (AME = 0.118, 95% CI [0.099, 0.137], $p<0.001$). Crucially, the GDPR awareness coefficients are virtually unchanged from M1: $\widehat{\beta}$ (Heard) moves from 0.637 to 0.633 and $\widehat{\beta}$ (Score) from 0.277 to 0.270.

The formal mediation analysis (Figure 3) confirms this pattern. For GDPR exposure (QB17), the indirect path is statistically non-significant (ACME = 0.003, 95% CI [-0.001, 0.006], $p = 0.096$), with 1.8% of the total effect mediated. For rights knowledge (QB18), the indirect path is statistically significant but negligible (ACME = 0.010, $p < 0.001$), accounting for only 3.0% of the total effect.

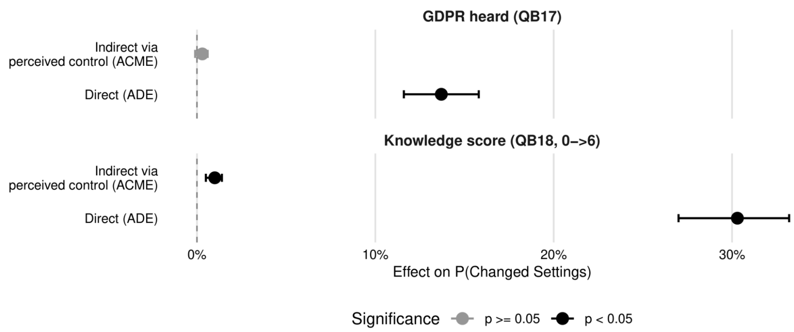

Figure 3. Mediation decomposition of the total effect of GDPR awareness on settings-changing behavior. ACME = average causal mediation effect (indirect via perceived control); ADE = average direct effect. 5,000 bootstrap simulations.

D. Robustness

Three robustness checks confirm the stability of the findings (Table 3). First, replacing the knowledge score with a binary indicator leaves the GDPR exposure coefficient unchanged ($\hat{\beta}$ = 0.633 vs. 0.637) and produces a large significant effect of any rights knowledge ($\hat{\beta}$ = 0.879, $p<0.001$). Second, substituting the country-level baseline from pre-GDPR perceived control to pre-GDPR privacy concern does not affect the GDPR awareness coefficients, though the baseline variable reverses sign. Third, excluding respondents who know all six rights (n = 721) yields a larger per-unit knowledge effect ($\hat{\beta}$ = 0.342 vs. 0.277). Across all specifications, GDPR awareness variables remain positive, large, and highly significant.

Table 3. Robustness checks

	M1: Main	**R1: Binary Knows**	**R2: Alt. Baseline**	**R3: Excl. Score=6**
(Intercept)	2.807***	2.998***	−0.716**	2.858***
	(0.334)	(0.335)	(0.270)	(0.340)
GDPR heard (QB17)	0.637***	0.633***	0.620***	0.647***
	(0.042)	(0.042)	(0.042)	(0.043)
Knowledge score (QB18)	0.277***		0.269***	0.342***
	(0.015)		(0.015)	(0.017)
Knows any right (binary)		0.879***		
		(0.042)		
Concern: data use (QB10)	−0.169***	−0.180***	−0.211***	−0.167***
	(0.026)	(0.026)	(0.027)	(0.027)
Concern: tracking (QB7)	−0.266***	−0.262***	−0.248***	−0.263***
	(0.020)	(0.020)	(0.020)	(0.020)
Country baseline control (2015)	−0.629***	−0.758***		−0.678***

	(0.140)	(0.141)		(0.143)
Country baseline concern (2015)			0.979***	
			(0.113)	
N	10,263	10,263	10,263	9,542
AIC	14,318.3	14,200.1	14,204.8	13,226.4

Note: Weighted binary logistic regression. Coefficients are log-odds. Standard errors in parentheses.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

E. Hypothesis Evaluation

The results unambiguously support Hypothesis 1 and reject Hypothesis 2. GDPR awareness, both passive exposure and substantive rights knowledge, has a strong positive effect on privacy-protective behavior across all model specifications. The institutional-trust suppression mechanism posited by H2 finds no empirical support: awareness coefficients are positive and stable even after controlling for perceived control, and there is no evidence that felt regulatory protection reduces behavioral engagement.

The mediation hypothesis embedded within H1 receives only partial support. While perceived control independently predicts behavior and GDPR awareness does predict felt control, the indirect path is too small to constitute a meaningful mechanism: 97% of the awareness–behavior association operates through direct pathways not captured by QB9. This suggests that regulatory literacy motivates privacy-protective action through channels beyond enhanced felt control - potentially including norm salience, rights activation, or behavioral self-efficacy, and points to the limits of perceived control as a mediating construct in regulatory contexts.

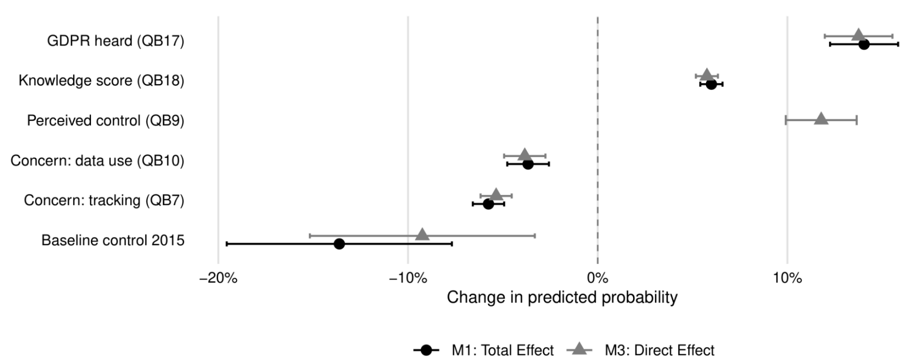

Figure 4. Average marginal effects on *P*(Changed Settings), comparing M1 (total effect) and M3 (direct effect, controlling for perceived control). 95% CIs.

V. Diagnostics

This section assesses potential threats to the validity of the logistic regression estimates across M1, M2, and M3, covering multicollinearity, separation, influential observations, and model fit.

A. Multicollinearity

Variance inflation factors (VIF) were computed for all predictors across the three models (Table 4). All VIF values fall below 2.0, well beneath the conventional thresholds of concern (VIF > 5) or severity (VIF > 10). The two GDPR awareness measures produce VIF values below 1.3, consistent with their low inter-correlation ($r = 0.20$; Table A4 in the Appendix).

Table 4. Variance Inflation Factors (VIF) across models

Predictor	M1: Total	M2: Mediator	M3: Full
GDPR heard (QB17)	1.021	1.044	1.022
Knowledge score (QB18)	1.022	1.055	1.023
Concern: data use (QB10)	1.007	1.008	1.009
Concern: tracking (QB7)	1.027	1.041	1.029
Country baseline control (2015)	1.013	1.012	1.025
Perceived control (QB9)	—	—	1.016

Note: All VIF values well below the threshold of concern (VIF > 5). No multicollinearity detected.

B. Complete separation

Binary logistic regression is vulnerable to complete separation, where a predictor or combination of predictors perfectly predicts the outcome, causing maximum likelihood estimates to diverge. Cross-tabulations of the binary predictors (QB17, QB18 binary) against the dependent variable (QB11) confirm that all cells are non-empty: there are respondents in every combination of awareness and behavior category. Complete separation is not present.

C. Heteroscedasticity

Heteroscedasticity in the classical OLS sense does not apply to binary logistic regression: the variance of a Bernoulli outcome is $p(1-p)$ by construction and this is incorporated directly into the likelihood function. The Pearson dispersion statistic for M1 is 1.10, close to 1.0, indicating no overdispersion.

D. Model fit

Table 2 reports AIC values for all three models. M3 (AIC = 14,158) achieves a modest improvement over M1 (AIC = 14,318) and M2 (AIC = 14,511). Pseudo-R^2 values are 0.075 for M1, 0.018 for M2, and 0.085 for M3. These values are modest, which is expected for cross-sectional survey data with a single behavioral outcome: the models capture meaningful variance in privacy-protective behavior, but a large share of individual-level behavioral variation remains unexplained by the included predictors – consistent with the privacy paradox literature's characterization of privacy behavior as context-dependent and difficult to predict from attitudinal variables alone.

VI. Conclusion

The surveillance economy operates on a structural asymmetry: companies possess both the incentive and the infrastructure to extract personal data at scale, while citizens are granted legal rights they must proactively exercise to resist it. The GDPR was designed to rebalance this relationship but legal architecture alone cannot close the governance gap if citizens lack the awareness to activate it. This paper has asked, does knowing about GDPR actually change what people do?

The answer, across three logistic regression models and a formal mediation analysis on 10,263 European citizens surveyed in 2019, the year following GDPR implementation, is affirmative and robust. Individuals who have heard of GDPR are 14.0 percentage points more likely to have changed their privacy settings than those who have not. Each additional data right known adds a further 6.0 percentage points to this probability, such that full regulatory literacy predicts a 36 percentage point behavioral advantage over complete ignorance. These findings hold across all robustness checks and are stable to alternative operationalizations of both the treatment and the baseline predictor.

Two secondary findings show that the privacy paradox is empirically visible in the same data: concern about data use and online tracking, namely the attitudinal variables that dominate prior survey research, are negatively associated with settings-changing behavior. It is regulatory knowledge, not concern, that predicts action. This distinction matters for policy: awareness campaigns targeting specific rights are likely to be more effective than those designed to heighten abstract concern.

Second, the mediation of the awareness–behavior relationship through perceived control is statistically detectable but substantively negligible. Only 3.0% of the effect of rights knowledge on behavior runs through enhanced felt control. GDPR awareness appears to motivate action through more direct pathways such as norm activation, rights salience, or behavioral self-efficacy, actions that question B9 does not fully capture. The paradox-of-control mechanism posited by Brandimarte et al. (2013) finds no support here: regulatory empowerment does not appear to generate complacency.

These findings contribute to the privacy paradox literature by demonstrating that a major regulatory intervention can narrow the attitude–behavior gap – but only where it produces genuine literacy and not exposure alone. Forty percent of those who have heard of GDPR cannot name a single right it guarantees. The governance gap this paper identifies is therefore not a failure of regulation, but of regulatory communication. Future research should examine the mechanisms through which rights knowledge is acquired and whether targeted legal literacy interventions can replicate the behavioral effects observed here at scale.

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Appendix A1: Sample composition by country

Table A1 reports country-level sample composition and key variable means for the analytic sample.

Country	*N*	Changed settings (%)	GDPR heard (%)	Mean knowledge score	Perceived control (%)
AT	553	64.0	49.5	1.43	72.3
BE	627	58.5	28.9	1.23	67.6
BG	359	48.7	33.4	1.13	56.3
CY	194	44.8	32.5	1.40	80.4
CZ	553	51.0	58.6	0.67	75.6
DE-E	196	47.4	51.5	1.45	55.6
DE-W	475	54.5	52.4	1.19	54.5
DK	585	70.8	61.9	1.26	65.1
EE	402	70.9	39.3	2.48	76.6
ES	475	63.8	47.6	1.05	59.4
FI	500	69.0	43.4	1.48	83.2
FR	452	66.2	21.5	1.22	66.8
GB	496	78.2	57.7	1.88	77.6
GR	434	52.3	48.8	1.06	71.0
HR	530	54.9	42.3	1.31	77.7
HU	435	45.7	34.5	1.10	71.3
IE	630	62.9	58.1	1.62	73.5
IT	567	46.6	25.0	1.21	68.6
LT	362	57.5	33.7	1.77	79.0
LU	270	66.3	47.8	1.15	64.4
LV	497	57.7	49.9	1.74	70.8
MT	172	72.1	58.7	1.86	79.7
NL	626	66.0	61.2	1.44	69.8
PL	372	53.5	70.7	1.45	84.1
PT	364	86.3	41.8	1.61	86.5
RO	418	50.5	43.3	1.44	67.9
SE	630	66.7	67.8	0.70	70.6
SI	438	62.6	42.9	0.89	70.8
SK	436	58.0	61.9	0.79	78.0

Note: Source: Special Eurobarometer 487a (EB 91.2, 2019). Post-stratification weights applied.

Appendix A2: Brant test for proportional odds assumption

Prior to estimating Model 2, an ordered logit was fitted with QB9 (perceived control, four categories) as the dependent variable. The Brant test was applied to assess whether the proportional odds assumption holds across all predictors. As reported in Table A3, the omnibus test rejects the assumption ($\chi^2 = 46.49$, df = 4, $p < 0.001$), with the violation driven by the concern about data use predictor (QB10). This justifies collapsing QB9 into a binary indicator and estimating Model 2 as a binary logit.

Variable	**χ^2**	***df***	***p*-value**
Omnibus	46.49	4	<0.001
GDPR heard (QB17)	1.70	1	0.193
Knowledge score (QB18)	0.34	1	0.557
Concern: data use (QB10)	44.11	1	<0.001
Concern: tracking (QB7)	0.11	1	0.744

Note: H_0: Parallel regression assumption holds. A significant result ($p < 0.05$) indicates a violation. The omnibus test failure driven by QB10 justifies collapsing QB9 into a binary outcome variable.

Appendix A3: Correlation matrix on GDPR awareness measures

Table A4 reports the correlation matrix for the two GDPR awareness measures. The low inter-correlation ($r = 0.20$) supports treating them as distinct predictors.

Variable	**GDPR heard (QB17)**	**Knowledge score (QB18)**	**Knows any right (binary)**
GDPR heard (QB17)	1.000	0.204	0.194
Knowledge score (QB18)	0.204	1.000	0.753
Knows any right (QB18, binary)	0.194	0.753	1.000

Note: Pearson correlations. N = 10,263. Low inter-correlation between QB17 and QB18 ($r = 0.20$) confirms the two measures capture empirically separable dimensions of GDPR awareness.

Note on the use of artificial intelligence

Model	**Usage**
GPT 5.4	• Used for final formatting *"Format this document into google doc"* • Used for reviewing the soundness of the paper. *"Do a final soundness review of this paper, flag any critical issues"* • Used to format the bibliography in the final document • Used to understand the bugs in my R code, namely: "*Why do I have this error here [pasted error]?*" • Used to prepare the R code to create the figures in the paper.